

Circles, Cyclicity & Tangency

Recognition chains before theorem selection

STRUCTURE → JUSTIFY → LOCAL RELATION → ROUTE → CHECK

Student learning pack · assimilation, recognition, transfer, mastery

Circles, Cyclicity and Tangency — Assimilation Book

A circle problem becomes much easier when you stop asking “Which theorem do I remember?” and start asking:

What circle structure is actually proved here, and what is the smallest local relation that can close the problem?

The working routine for this topic is:

IDENTIFY STRUCTURE → JUSTIFY IT → CHOOSE ONE LOCAL RELATION → CLOSE → CHECK

The picture may suggest a circle fact. The givens must justify it.

1. Reconnect: what you already own

You already know how to close ordinary angle information:

- angles on a line;
- angles in a triangle;
- parallel-line angle relations and their converses;
- the sum of the angles of a quadrilateral;
- symmetry only after it has been proved;
- coordinates as an alternate representation when they simplify the metric data.

Those facts remain available. They are **not** circle theorems.

The new question is:

What extra information does the circle add?

A generic quadrilateral gives only a 360° angle sum. A **cyclic** quadrilateral gives a much stronger local relation. A line that merely touches a drawing is not automatically a tangent. A product of four lengths is not automatically a power relation.

This topic is about earning the right to use the extra structure.

Part I — Angles created by a circle

2. The same chord is the first clue

Suppose A, B, C lie on a circle with centre O .

If $\angle ACB$ subtends chord AB , then the central angle subtending the same chord is twice as large:

$$\angle AOB = 2\angle ACB.$$

This is useful only when the two angles genuinely see the **same endpoints A, B** .

Recognition cue

Before writing an angle equation, underline the two endpoints of the chord or arc being viewed.

If two inscribed angles subtend the same chord AB from the same side of the chord, then they are equal.

So if C, D lie on the same arc-side of chord AB ,

$$\angle ACB = \angle ADB.$$

Why this matters

A complicated angle chase can often collapse as soon as you notice that two angles are simply different views of the same chord.

Contrast

Looks similar: two angles both have a point on the circle.

Not enough: they do not subtend the same chord.

The circle does not make arbitrary angles equal.

3. Diameter means a right angle — but only through the diameter

If AB is a diameter and C lies on the circle, then

$$\angle ACB = 90^\circ.$$

Why? The central angle $\angle AOB$ is 180° , so the inscribed angle is half of it.

First-move cue

If a problem gives a diameter, immediately scan for a triangle whose third vertex lies on the circle.

Common error

A long chord that looks like a diameter is not enough. You need the centre on the chord, or an explicit statement that it is a diameter.

Part II — Prove cyclicity before using cyclicity

4. Cyclic quadrilateral versus ordinary quadrilateral

For an ordinary quadrilateral $ABCD$,

$$\angle A + \angle B + \angle C + \angle D = 360^\circ.$$

If A, B, C, D are concyclic, then opposite angles are supplementary:

$$\angle A + \angle C = 180^\circ,$$

and

$$\angle B + \angle D = 180^\circ.$$

This is a stronger statement, so it needs a stronger hypothesis.

Recognition rule

Write a cyclic angle relation only after one of these has happened:

1. the problem explicitly says the four points are concyclic;
2. you have proved opposite angles supplementary;
3. you have proved two equal angles subtend the same segment;
4. another valid circle criterion has been established.

Converse as a proof tool

If, for a quadrilateral,

$$\angle A + \angle C = 180^\circ,$$

then the quadrilateral is cyclic.

This direction is often more useful than the theorem itself: you manufacture a 180° sum from ordinary angle facts, then **upgrade** the quadrilateral to cyclic.

The recognition chain

ordinary angles $\rightarrow$ supplementary pair $\rightarrow$ cyclicity $\rightarrow$ stronger circle relation.

Do not reverse the order.

5. Equal subtended angles can prove cyclicity

Suppose points A, B, C, D satisfy

$$\angle ACB = \angle ADB.$$

Both angles subtend segment AB . Under the usual non-degenerate configuration, this is a signal that A, B, C, D are concyclic.

This is especially effective when an angle chase naturally produces equality before the circle is visible.

Diagnostic contrast

Valid: prove the equal angles first, then claim cyclicity.

Invalid: draw a circle through the four points because it looks plausible, then use equal angles.

Part III — Tangency: a local perpendicular structure

6. Tangent and radius at the point of contact

If a line is tangent to a circle at T and O is the centre, then

$$OT \perp \text{tangent at } T.$$

This gives a right angle immediately.

Recognition cue

When you see the word **tangent**, locate all three of these objects:

- the centre;
- the contact point;
- the tangent line.

Then write the perpendicular relation before chasing any other angle.

Contrast: tangent versus chord

A chord meets the circle at two points. A tangent meets it at one point locally.

A line drawn nearly tangent is not a tangent theorem hypothesis.

7. Equal tangents from one external point

If tangents from external point P touch the same circle at A and B , then

$$PA = PB.$$

The hidden structure is two right triangles:

- $OA \perp PA$;
- $OB \perp PB$;
- $OA = OB$;
- OP is common.

So the two right triangles are congruent.

First-move cue

If the same external point sends two tangent segments to one circle, mark them equal before introducing variables.

Why not use this everywhere?

Equal tangents require the **same external point** and the **same circle**. Two unrelated tangent segments need not be equal.

Part IV — Alternate segment: translate a tangent angle into a circle angle**8. The theorem as a translation rule**

Let the tangent at A meet chord AB. If C is on the opposite arc, then the angle between the tangent and chord AB equals the inscribed angle subtending chord AB:

$$\text{angle}(\text{tangent at } A, AB) = \angle ACB.$$

The most useful way to remember this is not as a sentence. Remember the **shared chord AB**.

Recognition chain

tangent at A + chord AB → find an inscribed angle that also sees AB.

Contrast: alternate segment versus generic angle chase

If a tangent and chord appear together, first test whether alternate segment gives the target angle in one step.

Do not start a five-line parallel/triangle chase unless the circle relation fails to connect the needed endpoints.

Hypothesis discipline

You may use alternate segment only after tangency is stated or proved.

A line that merely touches the sketch is not enough.

Part V — Power of a point: one invariant, several surfaces**9. The idea before the formulas**

For a fixed circle and a fixed point P, different lines through P can produce the same length product. That common value is the power of P with respect to the circle.

The important learner question is:

Which point owns the power?

Do not begin by memorizing products. First identify P.

10. Intersecting chords inside a circle

Suppose chords AB and CD intersect at interior point P.

Then

$$PA \cdot PB = PC \cdot PD.$$

First-move cue

Circle the common intersection point P, then pair the two segments lying on each complete chord.

Wrong product warning

The product uses the two pieces of the **same line through P**.
Do not pair visually nearby segments from different chords.

11. Two secants from an external point

Suppose P is outside the circle. One secant meets the circle at near point A and far point B; another meets it at near point C and far point D.

Then

$$PA \cdot PB = PC \cdot PD.$$

Here PB and PD are whole distances from the external point to the far intersections.

Common error

Using “outside piece × inside piece” instead of
outside distance × whole secant distance.
Always label the four points in order from P before multiplying.

12. Tangent-secant power

If PT is tangent and PAB is a secant through the same external point P, then

$$PT^2 = PA \cdot PB.$$

Again, the owner is P.

Decision boundary

- two chords crossing inside -> chord product;
- two secants from outside -> external secant product;
- tangent plus secant from outside -> tangent square equals secant product.

The invariant is the same. The visible surface changes.

Part VI — Common chords and intersecting circles**13. Two circles intersecting at A and B**

If two circles with centres O_1, O_2 intersect at A, B, then the line joining the centres is perpendicular to the common chord AB.

At this level, use this as a local geometric fact; there is no need to build a full radical-axis chapter.

Recognition cue

Two circles + common chord + centres -> test perpendicularity.
This can combine with another right angle to prove two lines parallel or collinear.

Why this matters in hard problems

Sometimes the circle theorem does not directly give the numerical answer. It gives a **direction** or **alignment** that makes a coordinate or metric calculation short.

That is still a successful circle-first route.

Part VII — Choosing synthetic or coordinate geometry

14. Circle structure first; representation second

Coordinates are not a failure of synthetic geometry. They are an alternate representation.

Use this decision rule:

Prefer a short synthetic chain when

- the target is an angle;
- cyclicity/tangency immediately creates a local angle relation;
- a power product closes the required length relation;
- equal tangents remove variables.

Test coordinates when

- the diagram already has a right triangle, square or rectangle;
- lengths are numerical;
- symmetry gives a natural axis;
- the circle condition becomes one quadratic equation;
- a synthetic route would require several unrelated auxiliary constructions.

Bad coordinate use

Introducing eight variables before using the circle structure.

Good coordinate use

Use the circle relation to decide what needs to be preserved, then place the geometry so most coordinates are zero or symmetric.

Part VIII — Five historical patterns

The historical problems below are not five new theorems. They are five examples of **representation choice**.

15. Square inside a right triangle with a circle condition

The useful structure is not “square theorem plus circle theorem.”

It is:

1. place the right triangle on axes;
2. represent the square with one side length;
3. use line membership for one square vertex;
4. use the circle equation for the other;
5. reject the geometric branch that does not fit the interior configuration.

This pattern yields the verified historical value 29.

Transfer cue

When a circle is embedded in an otherwise coordinate-friendly metric configuration, use the circle condition as one precise equation rather than launching an angle chase.

16. Rectangle plus concyclicity

A rectangle alone gives many metric possibilities.

The extra statement that four points are concyclic is the decisive constraint.

The useful chain is:

rectangle coordinates $\rightarrow$ use equal lengths $\rightarrow$ impose cyclicity $\rightarrow$ exclude degenerate branch $\rightarrow$ finish metric ratio.

The canonical historical interpretation is non-degenerate; the branch that makes the stated cyclic condition vacuous is not accepted.

Verified historical value: 03.

17. Two circles tangent internally to a larger circle

The decisive local fact is that the common chord of the two inner circles is perpendicular to their line of centres.

A supplied right angle creates another perpendicular to the same chord. This aligns the centre geometry and turns the radius condition into a short invariant calculation.

Verified historical value: 10.

Transfer cue

When several circles interact, ask first what is forced about their centres and common chords before writing circle equations.

18. Symmetric isosceles triangle and a parallel chord

The problem has a circumcircle, but symmetry makes coordinates cheaper than a long chord theorem chain.

The efficient route is:

1. centre the base on an axis;
2. find the altitude;
3. locate the circumcentre on the symmetry axis;
4. use distance from the centre to the horizontal chord;
5. recover the chord length.

Verified historical value: 25.

19. Circumcentres inside a square

The word “circumcentre” is a signal for perpendicular bisectors.

In a unit square, coordinates make those bisectors cheap. The perimeter condition removes the remaining symmetric expression.

Verified historical value: 03.

Transfer cue

When several circumcentres occur in an orthogonal coordinate-friendly shape, compare coordinate cost with synthetic cost before committing.

Part IX — Error laboratory inside the lesson

20. “It looks cyclic.”

Error: writing opposite angles supplementary because four points appear to lie on a circle.

Repair: state or prove cyclicity first.

21. “The line touches the circle, so it is tangent.”

Error: reading tangency from a sketch.

Repair: use an explicit tangent statement, or prove the radius is perpendicular at the contact point.

22. “Power means multiply four lengths.”

Error: choosing products by visual proximity.

Repair: identify the power point, then pair segments along each line through that point.

23. “A circle problem must have a synthetic solution.”

Error: refusing coordinates even when the metric shape is already a right triangle, square or rectangle.

Repair: preserve the circle structure first, then choose the cheapest representation.

24. “Coordinates solve everything.”

Error: replacing a one-step tangent or cyclic angle relation with many variables.

Repair: test the shortest local circle theorem before coordinate expansion.

Part X — Adopt the method

Before doing any serious calculation, write these four lines in your head:

1. **Circle structure:** What is stated or proved — same chord, cyclicity, tangent, common chord, power point?
2. **Local relation:** What one theorem directly uses that structure?
3. **Representation:** Is a short synthetic chain cheaper, or do symmetry/metric givens make coordinates cheaper?
4. **Check:** Did I use a theorem whose hypotheses were actually established?

You have assimilated this topic when an unfamiliar diagram no longer makes you search randomly through circle theorems. You first identify the structure, justify it, and then choose the shortest legal move.

First-Step Reference — Circles, Cyclicity and Tangency

Use this only after you understand the ideas. It is a **routing sheet**, not a theorem catalogue.

The four-question start

Before calculation, ask:

1. **What circle structure is stated or proved?**
 2. **Which points/lines belong to that structure?**
 3. **What is the shortest legal relation?**
 4. **Would coordinates be cheaper after the structure is identified?**
-

If you see the same chord

For points on one circle:

- central angle on chord AB = twice an inscribed angle on chord AB;
- inscribed angles subtending the same chord from the same segment are equal;
- a diameter subtends a right angle.

First line: name the shared chord before writing an angle equation.

If four points may be cyclic

Use cyclic relations only after cyclicity is stated or proved.

Fast proof routes:

- opposite angles sum to 180° ;
- two suitable equal angles subtend the same segment.

Once cyclic:

- opposite angles are supplementary;
- same-chord angle relations become available.

Contrast: generic quadrilateral gives only total angle sum 360° .

If a tangent appears

At tangent point T with centre O:

$$OT \perp \text{tangent at } T.$$

From common external point P, tangent lengths to the same circle are equal:

$$PA = PB.$$

First line: mark the contact point and the radius perpendicular.

If tangent + chord appear

Test alternate segment before a long angle chase.

If tangent at A meets chord AB, then the tangent-chord angle equals the inscribed angle subtending AB in the opposite segment.

First line: write the chord whose endpoints both angles see.

If products of lengths appear

Identify the **power point P** first.

- chords AB, CD intersect inside at P: $PA \cdot PB = PC \cdot PD$;
- two secants from outside P: $PA \cdot PB = PC \cdot PD$ with near/far points ordered from P;
- tangent PT and secant PAB: $PT^2 = PA \cdot PB$.

First line: label the complete lines through P; do not multiply visually nearby segments.

If two circles intersect

Their line of centres is perpendicular to their common chord.

First line: identify the two centres and the common chord before writing metric equations.

Synthetic or coordinates?

Prefer synthetic first when

- the target is an angle;
- cyclicity or tangency gives a one- or two-step relation;
- one power equation closes the length target.

Test coordinates when

- there is a right triangle, square or rectangle;
- symmetry gives a natural axis;
- lengths are numerical;
- the circle condition becomes one simple equation.

Rule: circle structure first, representation second.

Seven danger signals

1. "It looks cyclic." -> prove cyclicity.
2. "It looks tangent." -> use stated/proved tangency only.
3. "Opposite angles supplementary." -> only after cyclicity.
4. "Those tangent lengths look equal." -> same external point, same circle.
5. "Multiply these four lengths." -> identify one power point first.
6. "Circle problem means synthetic." -> compare coordinate cost.
7. "Coordinates always work." -> test a one-step circle relation first.

Recognition and First-Line Lab

For each item, write only the **best first move** or choose the best option. Do not complete a long solution unless needed to justify the choice.

Recognition

- Four points A, B, C, D form a quadrilateral and the picture looks almost circular. Which statement is safe immediately? A. opposite angles are supplementary B. equal chords subtend equal angles C. the quadrilateral angle sum is 360° D. $ABCD$ is cyclic
- A, B, C, D lie on one circle and $\angle ACB$ and $\angle ADB$ both subtend chord AB . What relation should you test first? A. the angles are equal B. their sum is 180° C. both are 90° D. no relation
- AB is a diameter and C lies on the circle. Which local fact is immediately available? A. $AC = BC$ B. $\angle ACB = 90^\circ$ C. $\angle ABC = 45^\circ$ D. $AB = AC$
- A line is stated to be tangent at T to a circle with centre O . What should be marked first? A. OT equals the tangent length B. OT is perpendicular to the tangent C. T is a midpoint D. the tangent is a chord
- From an external point P , two tangents touch the same circle at A, B . Which relation is cheapest? A. $PA = PB$ B. $OA = PA$ C. AB is a diameter D. $OP = OA$
- A tangent at A and chord AB form an angle. Which theorem should be tested before a generic angle chase? A. Pythagoras B. alternate segment C. quadrilateral sum D. inclusion-exclusion
- Chords AB and CD intersect at interior point P . Which product has the correct structure? A. $PA \cdot PC = PB \cdot PD$ B. $PA \cdot PB = PC \cdot PD$ C. $AB \cdot CD = PA \cdot PD$ D. $PA^2 = PC \cdot PD$
- Two circles intersect at A, B . What line relation should you inspect if their centres are known? A. line of centres is parallel to AB B. line of centres is perpendicular to AB C. all four points are collinear D. centres lie on the circles

First lines

- In cyclic quadrilateral $ABCD$, $\angle A = 112^\circ$. Write the first line that determines $\angle C$.
- A quadrilateral has $\angle A + \angle C = 180^\circ$. Write the first structural conclusion.
- A tangent at T to a circle with centre O meets a chord through T . Write the first angle relation you would mark before using alternate segment.
- External point P has tangent PT and secant meeting the circle at A then B . Write the first power equation.
- Two secants from external point P meet the circle in order $P-A-B$ and $P-C-D$. Write the correct power equation.
- A right triangle, a square and a circle condition all appear with numerical side lengths. Write one sentence explaining when coordinates should be tested.
- A proposed solution says “ $ABCD$ is cyclic because it looks cyclic.” Write the missing proof target, not a full proof.
- Two intersecting circles share chord AB ; another line through point A is also known perpendicular to AB . State the structural comparison you should make first.

Practice and Transfer Bank

Try each stage without looking ahead. The support fades as you move down the page.

Reconnect

1. A central angle subtending chord AB is 128° . Find an inscribed angle subtending the same chord from the corresponding arc.
2. $ABCD$ is cyclic and $\angle A = 104^\circ$. Find $\angle C$.
3. A line is tangent at T to a circle with centre O . Find the angle between OT and the tangent.
4. From external point P , tangents PA, PB touch the same circle. If $PA = 13$, find PB .

Structure with support

5. Points A, B, C, D lie on one circle. Angles $\angle ACB$ and $\angle ADB$ subtend chord AB from the same segment. If $\angle ACB = 37^\circ$, find $\angle ADB$.
6. A convex quadrilateral has opposite angles 72° and 108° . What circle-structure conclusion is available?
7. AB is a diameter of a circle of length 10, and C lies on the circle. If $AC = 6$, find BC .
8. From external point P , PT is tangent and secant PAB meets the circle at A then B . If $PT = 12$ and $PA = 8$, find PB .

Reduced support

9. Chords AB and CD intersect at P inside a circle. If $PA = 3$, $PB = 12$ and $PC = 4$, find PD .
10. Two secants from external point P meet the circle in order $P-A-B$ and $P-C-D$. If $PA = 5$, $PB = 20$ and $PC = 4$, find PD .
11. A tangent at A forms an angle of 48° with chord AB . Point C lies on the opposite arc. Find $\angle ACB$.
12. From point P , tangents touch a circle at A, B . If $PA = 10$ and $AB = 12$, find the perimeter of triangle PAB .

Mixed independent

13. Prove that tangent segments from the same external point to a circle have equal lengths. Your proof must identify the two right triangles used.
14. In a convex quadrilateral $ABCD$, suppose $\angle ABD = \angle ACD$. Explain why this is a cyclicity signal and state the conclusion you would justify before using any cyclic-quadrilateral theorem.
15. Two circles intersect at A, B and have centres O_1, O_2 . A line through A is also perpendicular to AB . State the direction relation this gives with O_1O_2 , and explain why.
16. A problem contains a rectangle, two side points, two equal-length conditions and a concyclicity condition, and asks for a side ratio. Explain why a coordinate placement may be cheaper than a long synthetic chase, but name the circle condition that must still be imposed explicitly.

Transfer

17. Historical anchor — symmetric isosceles circumcircle.

An isosceles triangle has sides 20, 20, 30. The altitude from the apex is bisected at M . A chord of the circumcircle through M is parallel to the base. Determine the chord length.

18. Historical anchor — square/right-triangle/circle metric.

A right triangle has legs 1 and 2. A square sits on the longer leg, one upper vertex lies on the hypotenuse, and the other upper vertex lies on the circle centred at the remaining acute vertex and passing through the right-angle vertex. If the square area is m/n in lowest terms, determine $m+n$.

19. Historical anchor — rectangle plus cyclicity.

A rectangle has points M, N on two adjacent sides with $MC = CD$, $MD = MN$, and C, D, M, N concyclic. Under the canonical non-degenerate reading, $(AB/BC)^2 = m/n$ in lowest terms. Determine $m+n$.

20. Historical anchor — internally tangent circles.

A circle of radius 10 contains two circles tangent to it internally. The two inner circles intersect at A, B , and the radius from the large centre to A is perpendicular to the common chord AB . Determine the sum of the two inner radii.

21. Historical anchor — two circumcentres in a unit square.

In a unit square, points M, N lie on consecutive sides near the same vertex, with the perimeter of the small corner triangle equal to 2. Let O be the circumcentre of one triangle involving M, A, N , and P the circumcentre of the triangle involving M, O, N . If $(OP/OA)^2 = m/n$ in lowest terms, determine $m+n$.

22. Author-created mixed power transfer.

From external point P , tangent PT and two secants PAB, PCD meet the same circle. Suppose $PA = 4$, $PB = 9$ and $PC = 6$. Determine both PD and PT , and state the single invariant that makes the two answers agree.

Benchmark Assimilation Lab — Circles, Cyclicity and Tangency

Use this after the Assimilation Book and before the final mastery test. The aim is to test whether you can **recognize and justify circle structure before selecting a theorem**.

Learning loop:

RECONNECT -> DIAGNOSE -> ADOPT -> TRANSFER

A. RECONNECT — no notes

Write only the first useful relation unless a short explanation is requested.

1. What must be checked before using opposite supplementary angles in a quadrilateral?
2. If two inscribed angles see the same chord from the same segment, what relation do you test first?
3. What local right angle comes from a stated tangent?
4. What must be common before two tangent segments can be declared equal?
5. In a tangent-chord configuration, what geometric object links the tangent angle to an inscribed angle?
6. Before writing a power product, what point must be identified?
7. What relation connects the centres of two intersecting circles to their common chord?
8. When should coordinates be tested instead of extending a synthetic angle chase?

Diagnostic interpretation:

- misses 1-2 -> cyclicity/same-chord recognition is weak;
- misses 3-5 -> tangency/alternate-segment bridge is weak;
- misses 6 -> power-point ownership is weak;
- misses 7 -> intersecting-circle structure is weak;
- misses 8 -> representation-selection boundary is weak.

B. Error laboratory

For each proposed solution, identify the first invalid or inefficient move and repair it.

Error 1 — cyclic from appearance

Claim: "The four points look as if they lie on one circle, so opposite angles sum to 180° ."

Repair target: state what must be proved or given first.

Error 2 — tangent from appearance

Claim: "The line touches the circle in the drawing, so the radius is perpendicular to it."

Repair target: require stated tangency or a valid proof of tangency.

Error 3 — wrong power pairing

Claim: "Two chords cross at P, so multiply the two shortest pieces and set that equal to the product of the two longest pieces."

Repair target: pair the two segments on each complete line through P.

Error 4 — wrong secant length

Claim: "For an external secant, power is outside piece times inside piece."

Repair target: use distance from the external point to the near intersection times distance from the external point to the far intersection.

Error 5 — theorem search before structure

Claim: "There is a circle, so list all circle theorems and see which one contains the target."

Repair target: identify the stated/proved structure and the smallest local chain first.

Error 6 — synthetic loyalty

Claim: “Because the problem is about a circumcircle, coordinates are not allowed.”

Repair target: compare representation cost after preserving the circle condition.

C. ADOPT — first move only

Do not solve fully. Write the first two useful lines.

1. A cyclic quadrilateral has one angle 117° .
2. A convex quadrilateral has opposite angles 83° and 97° .
3. A tangent at T is drawn to a circle with centre O, and a chord through T is also present.
4. From external point P, two tangents touch the same circle.
5. Chords AB and CD intersect at interior point P.
6. Tangent PT and secant PAB come from the same external point.
7. Two circles intersect at A, B; their centres are known.
8. A symmetric isosceles triangle has a circumcircle and a chord parallel to the base through a point on the symmetry axis.

D. TRANSFER — changed surface

1. A quadrilateral problem never mentions a circle, but an angle chase gives one pair of opposite angles summing to 180° . Explain the representation upgrade and what new relations become legal.
2. A road touches a circular park at T, and two measured paths from an external point meet the park at different places. Translate the useful tangent/secant information into one power statement.
3. Two sensors are centres of overlapping circular ranges. Their two boundary intersections are A, B. Explain the direction information supplied by the common chord before any coordinates are introduced.
4. A rectangular coordinate configuration contains four concyclic points and two equal-length constraints. Explain why cyclicity is not decorative information and where it should enter the algebra.
5. A diagram contains a tangent and many ordinary triangle angles. Explain how you decide whether alternate segment is cheaper than a generic angle chase.
6. Construct a problem where a circle is present but the cheapest solution is coordinate geometry. State the circle relation that must still be preserved explicitly.

E. Six-question assimilation test

Choose one problem from Section D and answer all six questions before completing its solution.

1. **What did you notice?** State the circle structure that matters.
2. **Why is that structure legal?** State the hypothesis or proof that earns it.
3. **What clue would make you think of it again?** Give a future recognition cue.
4. **What similar-looking situation needs a different method?** State one contrast pair.
5. **Can you write the first two useful lines without help?** Write them.
6. **Can you solve a disguised version?** Change the surface while preserving the same circle invariant and outline the route.

Readiness rule

You are not ready merely because you can recite circle theorems. You are ready when you can refuse an unjustified cyclic/tangent claim, identify the chord or power point that owns the relation, choose a short synthetic chain when one exists, switch to coordinates when the metric structure makes them cheaper, and start a disguised circle problem without a chapter label.

Mixed Mastery Test

Solve each problem without method labels or hints. Use only circle relations whose hypotheses are stated or proved.

1. In a circle with centre O , chord AB subtends central angle 146° . Point C lies on the corresponding arc-side so that $\angle ACB$ subtends the same chord. Determine $\angle ACB$.
2. $ABCD$ is cyclic and $\angle B = 68^\circ$. Determine $\angle D$.
3. A convex quadrilateral has $\angle A = 95^\circ$ and $\angle C = 85^\circ$. State the strongest circle-structure conclusion justified by this information.
4. From external point P , tangent PT and secant PAB meet the same circle. If $PT = 15$ and $PA = 9$, determine PB .
5. Chords AB and CD intersect at P inside a circle. If $PA = 6$, $PB = 8$ and $PC = 4$, determine PD .
6. Two secants from external point P meet a circle in order $P-A-B$ and $P-C-D$. If $PA = 3$, $PB = 18$ and $PC = 6$, determine PD .
7. A tangent at A forms an angle of 41° with chord AB . Point C lies on the opposite arc. Determine $\angle ACB$.
8. From an external point P , tangents touch a circle at A, B . Suppose $PA = 17$ and $\angle APB = 60^\circ$. Determine AB .
9. Two circles with distinct centres O_1, O_2 intersect at A, B . Prove that O_1O_2 is perpendicular to the common chord AB .
10. A circle has equation $x^2 + y^2 = 25$. A horizontal chord lies on the line $y = 3$. Determine the chord length, and explain why coordinates are a natural route here.
11. A sketch shows line l touching a circle at T , but the statement does not say l is tangent and gives no right-angle information. A proposed solution writes $OT \perp l$, where O is the centre. Explain precisely why this move is invalid and what would be sufficient to justify it.
12. A tangent at A and chord AB form an angle of 52° . Points B, C, D, A lie on one circle, with C in the opposite segment from the tangent angle. Determine $\angle ACB$, then state one additional cyclic relation involving an angle at D that also subtends chord AB from the same segment as C .